

Adil NAS

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Summary

Honors graduate and M.Sc. student in Computer Engineering at Sivas Cumhuriyet University, specializing in robotics, autonomous systems and software development. Led my university team to 2nd place in Turkey at Teknofest with a fully autonomous robot, personally owning the ROS2 software, SLAM, navigation and sensor integration. Currently leading the same team on an autonomous forklift AGV for in-factory logistics (PLC/automation communication, mapping, route following, computer vision and QR-based localization). Completed AI and machine-learning training at Samsung Innovation Campus. Experienced in embedded systems and full-stack development, with publications on autonomous robotics and software defect prediction.

Publications

NAS, A.; Kekül, H.; Yüksek, A. G. "Comparative Analysis of Data Balancing and Dimensionality Reduction Strategies in Software Defect Prediction." ICIAS 2026 (International Conference). [View Paper](#)

NAS, A.; Mandacı, E. D.; Yüksek, A. G. "IoT-Enhanced MIS Architectures for Autonomous Robotic Systems." Information Technologies in Managerial Decision-Making: Foundations and Application (Book Chapter), 2025. [DOI](#)

Education

Cumhuriyet University — Computer Engineering, M.Sc.

GPA: 3.75 / 4.00

Sivas

Sep 2025 – Present

Thesis: Autonomous Mobile Robot Navigation System Based on Visual, Auditory and Spatial Sensor Data Fusion

Cumhuriyet University — Computer Engineering, B.Sc.

GPA: 3.03 / 4.00 – Honor Student

Sivas

Sep 2021 – Jun 2025

Experience

Teknofest — Robotics & AI Team Lead

Sivas

Sep 2024 – Present

- Won 2nd place in Turkey in the 2025 Industrial Digital Technologies competition (Advanced Category) with a fully autonomous robot.
- Led the team and developed all ROS2-based software, including sensor integration, SLAM (SLAM Toolbox) and autonomous navigation (Nav2); contributed to hardware and electronics.
- Currently (2026) building an autonomous forklift AGV for in-factory logistics: mapping, route following, obstacle detection, PLC/factory-automation communication, vision-based line following and QR-code localization.
- Owned the full software pipeline end to end; gained deep experience in robotics, sensor fusion and real-world autonomous systems.

Samsung Innovation Campus — AI Intern

Remote

Jul – Nov 2025

- Completed a structured AI program covering machine-learning and deep-learning fundamentals.
- Performed data analysis, model development and evaluation in Python.
- Gained hands-on experience with LLM orchestration tools such as Ollama, CrewAI and LangChain.
- Built and presented an end-to-end AI capstone project.

C-Prot — Software Intern

Mersin

Jul – Aug 2024

- Developed file-scanning and malware-detection algorithms in C++.
- Built a project focused on verifying the digital certificate status of executable files.
- Developed a full-stack application in Tauri using Svelte and Rust.

Projects

Multi-Robot Autonomous Warehouse System — ROS2 Humble, Nav2, Gazebo, Docker

- Built a centrally-managed system where four AGVs autonomously execute pick-and-drop tasks in a Gazebo environment.
- Implemented conflict-free path planning with Conflict-Based Search (CBS) + A*, and task allocation via Hungarian/greedy algorithms, plus battery and auto-charging management.
- Designed a modular architecture of 6 ROS2 packages with a real-time Flask/SocketIO web monitoring dashboard.

ROS2 Differential-Drive Hardware Interface — ros2_control, C++, Nav2

- Developed a ros2_control SystemInterface serial bridge for motor/encoder control, integrated with the full Nav2 stack (SLAM, AMCL, EKF sensor fusion).
- Designed it for Arduino Mega and STM32F407 boards across ROS2 Humble/Jazzy distributions.

Autonomous Line-Following Robot — Raspberry Pi, OpenCV, PID

- Built an autonomous robot that processes PiCamera frames and adjusts motor speeds via PID control based on contour-based line position and angle.

Security & Browser Tools — Chrome MV3, JavaScript

- Developed a production-grade ad blocker (220+ rules), an HTTP request-analysis/DevTools extension, and cookie/JWT security-analysis tools.

Technical Skills

Programming Languages: C++, Python, JavaScript, C#, Java, HTML/CSS, SQL

Robotics & AI: ROS2, Nav2, SLAM Toolbox, ros2_control, Gazebo, OpenCV, Ollama, CrewAI, LangChain

Technologies & Tools: React, Node.js, Flask, Tauri, Svelte, Docker, Linux, Git/GitHub, Arduino, Raspberry Pi, Jetson, STM32

Protocols & Interfaces: I2C, SPI, UART/USART, CAN, PWM, GPIO, PLC communication

Certifications

- Teknofest 2025 – Industrial Digital Technologies, Turkey 2nd Place Certificate · TEKNOFEST (2025)
- Samsung Innovation Campus · Samsung Electronics (2025)
- ROS2 | Advanced ROS2: Modeling, Nav2 & SLAM · Udemy (2025)
- OpenCV Bootcamp · Python for Beginners · OpenCV University (2026)
- TEI Aircraft Engines School · TEI / TUSAŞ Engine Industries (2026)
- Version Control: Git & GitHub · Database Attacks and Database Security · BTK Academy (2024)

Languages

- **English:** Intermediate (B1–B2)
- **Turkish:** Native

Soft Skills

Fast Learning & Problem Solving • Team Leadership & Collaboration • Strong Research Skills • Eager to Learn New Technologies

Interests

Robotics, Autonomous Systems, Embedded Systems, Artificial Intelligence, Web Technologies, Open-Source Projects

References

Available upon request.